

1) 4-14

$$J(x) = \int_0^{t_f} \left[ \frac{\sqrt{1+\dot{x}^2(t)}}{x(t)} \right] dt$$

$x(0) = 0$ ,  $x(t_f)$ 는  $\theta(t) = t-5$  위에 있어야 한다.

E-L을 이용하면

$$\frac{d}{dt} \left( \frac{1}{x} \cdot \dot{x} \cdot \frac{1}{2} (1+\dot{x}^2)^{-\frac{1}{2}} \right) - \left( (1+\dot{x}^2)^{-\frac{1}{2}} (-1) x^{-2} \right) = 0$$

$$\frac{d}{dt} \left( \underbrace{(x^{-1} \dot{x} (1+\dot{x}^2)^{-\frac{1}{2}})}_A - \underbrace{(1+\dot{x}^2)^{-\frac{1}{2}} (-1) x^{-2}}_B \right) = 0$$

$$\Rightarrow \underbrace{-(x^{-1})^2 \dot{x}^2 (1+\dot{x}^2)^{-\frac{1}{2}}}_B + \underbrace{(x^{-1})^{-1} \ddot{x} (1+\dot{x}^2)^{-\frac{1}{2}}}_B + \underbrace{(x^{-1})^{-1} \dot{x} (-\frac{1}{2})(1+\dot{x}^2)^{-\frac{3}{2}} \cdot (2\dot{x})}_{A \cdot (\ddot{x})} + \underbrace{(x^{-1})^2 (1+\dot{x}^2)^{\frac{1}{2}}}_B = 0$$

$$\Rightarrow \left[ (1+\dot{x}^2)^{-\frac{1}{2}} \left\{ -\frac{\dot{x}^2}{x^2} + \frac{\ddot{x}}{x} \right\} + (1+\dot{x}^2)^{-\frac{3}{2}} \left\{ -\frac{\dot{x}^2}{x} \dot{x} \right\} + (1+\dot{x}^2)^{\frac{1}{2}} \left( \frac{1}{x^2} \right) \right] = 0$$

$$\Rightarrow \left[ (1+\dot{x}^2) \left\{ -\frac{\dot{x}^2}{x^2} + \frac{\ddot{x}}{x} \right\} - \left\{ \frac{\dot{x}^2}{x} \ddot{x} \right\} + \left( \frac{1}{x^2} \right) (1+\dot{x}^2)^2 \right] = 0 \quad \begin{matrix} \times (1+\dot{x}^2)^{\frac{3}{2}} \\ \text{양변에 곱함} \end{matrix}$$

$$\Rightarrow (1+\dot{x}^2) \left\{ -\dot{x}^2 + \ddot{x} \cdot x \right\} - \left\{ \dot{x}^2 \ddot{x} x \right\} + (1+\dot{x}^2)^2 = 0 \quad \begin{matrix} \times x^2 \end{matrix}$$

$$\Rightarrow -\dot{x}^2 - \cancel{\dot{x}^4} + \ddot{x} x + \cancel{\dot{x}^2 \ddot{x} x} - \cancel{\dot{x}^2 \ddot{x} x} + 1 + 2\dot{x}^2 + \cancel{\dot{x}^4} = 0$$

$$\Rightarrow +\dot{x}^2 + \ddot{x} x + 1 = 0$$

$$\therefore \frac{d}{dt} (\dot{x}^2 + \ddot{x} x) = -1 \quad \text{이므로 적분하여 정리하면}$$

$$x \dot{x} = -t + C_1 \quad \text{이고, 이것을 더 적분하면} \quad \frac{1}{2} x^2 = -\frac{1}{2} t^2 + C_1 t + C_2$$

주어진 조건에 의해  $x(0) = 0$  이므로  $C_2 = 0$ .

또한, Boundary Condition에 의해

$$\Rightarrow -t_f^2 + C_1 t_f = t_f^2 - 10t_f + 25$$

$$\text{Transversality Condition에 의해} \quad g + \frac{\partial g}{\partial \dot{x}} [\dot{\theta} - \dot{x}] = 0$$

$$\frac{\sqrt{1+\dot{x}^2}}{x} + \frac{1}{x} \left( \frac{1}{2} \right) (2\dot{x}) (1+\dot{x}^2)^{-\frac{1}{2}} (1-\dot{x}) = 0$$



$$\frac{\sqrt{1+\dot{x}^{*2}}}{x^*} + \frac{\dot{x}^*}{x^*} \frac{\sqrt{1+\dot{x}^{*2}}}{1+\dot{x}^{*2}} (1-\dot{x}^*) = 0$$

$$1 + \frac{1-\dot{x}^*}{1+\dot{x}^{*2}} \dot{x}^* = 0$$

$$1 + \dot{x}^{*2} + \dot{x}^* - \dot{x}^{*2} = 0$$

$$\therefore \dot{x}^*(t_f) = -1$$

$$2t_f^2 - (C_1 + 10)t_f + 25 = 0 \quad (\text{앞의 B.C 식}) \quad \text{이고,}$$

$$x^*(t) = \sqrt{-t^2 + C_1 t} \quad \text{임을 앞에서 구했으므로}$$

$$\dot{x}^*(t) = (-2t + C_1)(-t^2 + C_1 t)^{-\frac{1}{2}} = -1$$

$$\therefore (-2t + C_1)^2 (-t^2 + C_1 t)^{-1} = 1$$

$$-t_f^2 + C_1 t_f = (-2t_f + C_1)^2 = 4t_f^2 - 4C_1 t_f + C_1^2$$

$$5t_f^2 - 5C_1 t_f + C_1^2 = 0$$

$$\therefore t_f = 5 \pm \sqrt{\frac{25}{2}}, \quad C_1 = 10 \Rightarrow x^{*2}(t) = -t^2 + 10t \quad x^*(t) = \sqrt{-t^2 + 10t}$$

2) 4-16

$$E-L: x^{*2}(0) = 0 \rightarrow C_2 = 0$$

$$x^{*2}(t_f) = 9 - (t_f - 9)^2$$

$$\text{tr. cond} : g(x, t_f) + \frac{\partial g}{\partial \dot{x}}(x, t_f) [\dot{\theta}(t_f) - \dot{x}^*(t_f)] = 0$$

$$\frac{\sqrt{1+\dot{x}^{*2}(t_f)}}{x^*(t_f)} + \frac{\dot{x}^*(t_f)}{x^*(t_f)\sqrt{1+\dot{x}^{*2}(t_f)}} \times \left[ \frac{-(t_f - 9)}{x^*(t_f)} - \dot{x}^*(t_f) \right] = 0$$

$$(1 + \dot{x}^{*2}(t_f)) + \dot{x}^*(t_f) \left[ \frac{-(t_f - 9)}{x^*(t_f)} - \dot{x}^*(t_f) \right] = 0$$

$$x^*(t_f) - \dot{x}^*(t_f)(t_f - 9) = 0$$



B.C 에 의해  $2x^*(t_f)\dot{x}^*(t_f) = -2t + C$  이고

앞의 페이지 마지막 식에 의해

$$9 - (t_f - 9)^2 - (-t_f + \frac{C_1}{2})(t_f - 9) = 0$$

$$9 - (t_f^2 - 18t_f + 81) - (-t_f + (9 + \frac{C_1}{2})t_f - \frac{9}{2}C_1) = 0$$

$$(9 - \frac{C_1}{2})t_f + (-72 + \frac{9}{2}C_1) = 0 \quad \text{를} \quad \text{맨} \quad \text{윗줄} \quad \text{식에} \quad \text{대입하면}$$

$$\therefore C_1 = 8, \quad t_f = \frac{36}{5} = 7.2 \quad \Rightarrow \quad x^{*2}(t) = -t^2 + 8t$$

3)

교재를 참고하여

$$\begin{cases} \dot{\vec{x}}^*(t) = \frac{\partial H}{\partial \vec{p}}(x, t) \\ \dot{\vec{p}}^*(t) = -\frac{\partial H}{\partial \vec{x}}(x, t) \\ 0 = \frac{\partial H}{\partial u}(x, t) \end{cases} \quad \begin{aligned} H &= \frac{1}{2}(x_1^2 + x_2^2 + u^2) + p_1(t)x_2(t) + p_2(t)\{-x_1(t) - x_2(t) + u(t)\} \\ \dot{p}_1^*(t) &= -\frac{\partial H}{\partial x_1} = -(x_1^*(t) - p_2^*(t)) \\ \dot{p}_2^*(t) &= -\frac{\partial H}{\partial x_2} = -(x_2^*(t) + p_1^*(t) - p_2^*(t)) \\ 0 &= \frac{\partial H}{\partial u} = u^*(t) + p_2^*(t) \rightarrow u^*(t) = -p_2^*(t) \end{aligned}$$

$$\dot{x}_1^*(t) = x_2^*(t)$$

$$\dot{x}_2^*(t) = -x_1^*(t) - x_2^*(t) - p_2^*(t)$$

$$\begin{pmatrix} \dot{x}_1^*(t) \\ \dot{x}_2^*(t) \\ \dot{p}_1^*(t) \\ \dot{p}_2^*(t) \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & -1 & 0 & -1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 1 \end{pmatrix}}_A \begin{pmatrix} x_1^*(t) \\ x_2^*(t) \\ p_1^*(t) \\ p_2^*(t) \end{pmatrix} \Rightarrow \begin{pmatrix} x_1^*(t) \\ x_2^*(t) \\ p_1^*(t) \\ p_2^*(t) \end{pmatrix} = e^{At} \begin{pmatrix} x_1^*(0) \\ x_2^*(0) \\ p_1^*(0) \\ p_2^*(0) \end{pmatrix}$$

$$t=1 \rightarrow e^{At} = \begin{pmatrix} 0.5858 & 0.6524 & 0.1663 & -0.4972 \\ -0.8187 & 0.4306 & 0.4912 & -0.9833 \\ -0.4845 & -0.8282 & 0.2533 & 1.8131 \\ 1.1607 & -0.8171 & -1.6468 & 2.3973 \end{pmatrix}$$



1a)

$$\vec{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{x}(1) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$0 = 0.5858 + 0.1663 p_1^*(0) - 0.4972 p_2^*(0)$$

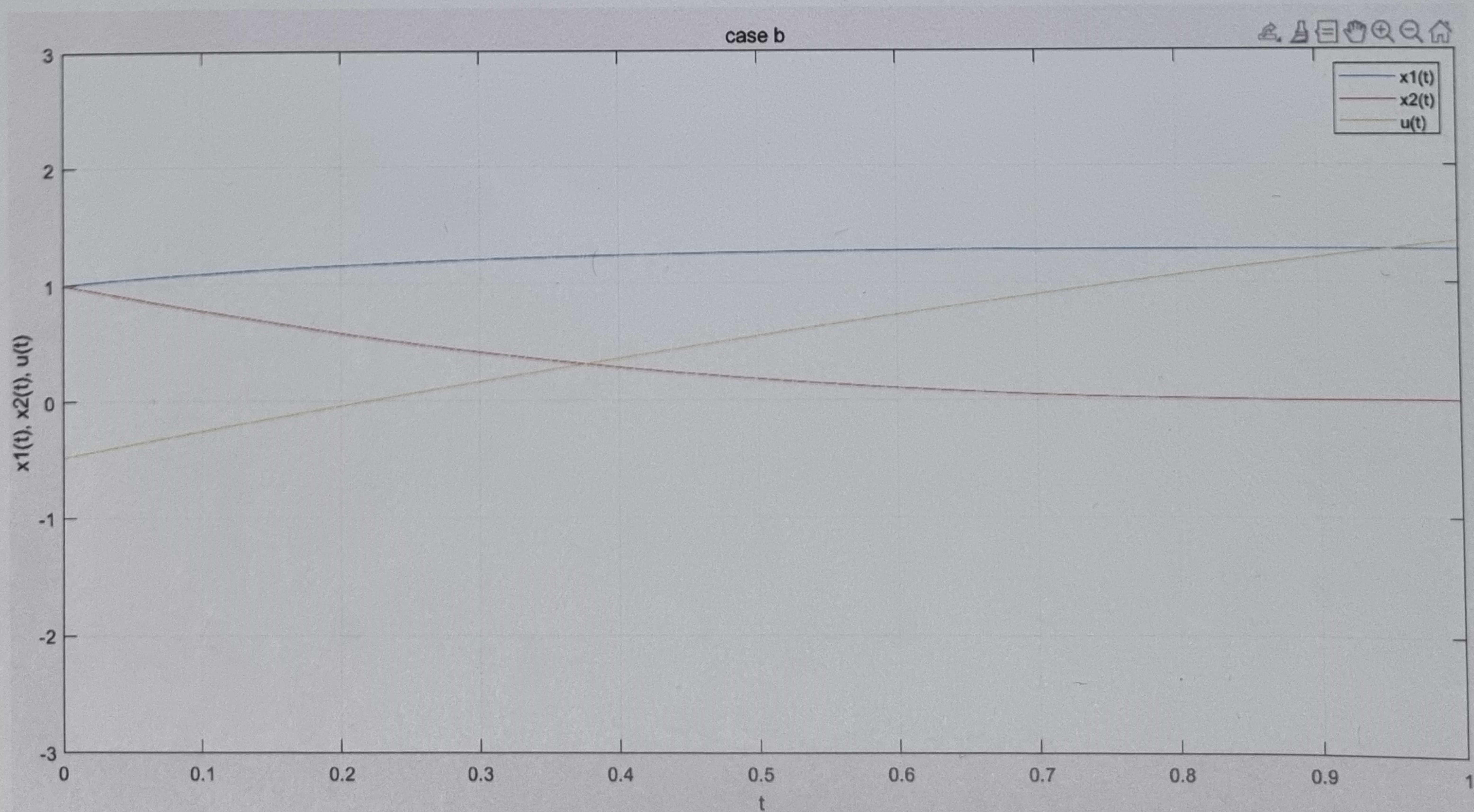
$$0 = -0.8187 + 0.4972 p_1^*(0) - 0.9833 p_2^*(0)$$

$$\rightarrow p_1^*(0) = 11.7473, \quad p_2^*(0) = 5.10736$$

$$p_1^*(1) = 11.7509, \quad p_2^*(1) = -5.9411$$

$$u^*(0) = -p_2^*(0) = -5.10736$$

$$u^*(1) = -p_2^*(1) = 5.9411$$





1b)

$$\vec{x}(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \vec{x}(1) = \begin{pmatrix} \text{free} \\ 0 \end{pmatrix}$$

$$\frac{\partial h}{\partial \vec{x}}(x, t_f) - \vec{p}^*(t_f) = 0$$

$$x_1 : x_1^*(t_f) - p_1^*(t_f) = 0 \rightarrow x_1^*(0), p_1^*(1) = \text{free}$$

$$x_2 : x_2^*(t_f) - p_2^*(t_f) = 0 \rightarrow x_2^*(1) = p_2^*(1) = 0$$

$$x_2^*(1) = 0 = -0.8187 + 0.4306 + 0.4972 p_1^*(0) - 0.9833 p_2^*(0)$$

$$p_2^*(1) = 0 = 1.1607 - 0.8171 - 1.6468 p_1^*(0) + 2.3973 p_2^*(0)$$

$$\rightarrow p_1^*(0) = -1.38649, \quad p_2^*(0) = -1.09576$$

$$p_1^*(1) = -3.6506 \quad p_2^*(1) \approx 0 = x_2^*(1)$$

$$u^*(0) = -p_2^*(0) = 1.09576$$

$$u^*(1) = -p_2^*(1) \approx 0$$

